Length Is the Floor: A Length-Residualized Audit of Training-Free Dynamic-RAG Triggers

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Abstract

Dynamic retrieval-augmented generation systems decide *when* to retrieve from a scalar trigger signal, scored by raw AUROC against binary correctness. Recent work has shown this protocol is length-confounded on adjacent QA selective-prediction and uncertainty-quantification (UQ) tasks. We present the first length-residualized audit of the training-free dynamic-RAG trigger pool on a paired-oracle target (`lift_em`$= +1$) that asks whether retrieval flips a wrong answer to a right one, isolating the causal retrieval-helpfulness signal from length-correlated incorrectness. Across two open-weight chat models on HotpotQA and 2WikiMultiHopQA (n=800 paired traces), nine signals beat the length baseline at 95% paired-bootstrap and only four survive Bonferroni at k=15. On the held-out 2Wiki cells, raw length collapses to at-or-below chance (AUROC ≤ 0.52), and our **FCT** (first-commitment time, a forward-looking answer-trajectory signal) reaches raw AUROC **0.784** on Gemma $\times$ 2Wiki, the strongest length-orthogonal signal in the pool. The survivor set is robust to a GPT-4o-Mini semantic-equivalence judge ($\kappa = 0.707$), and FCT itself *strengthens* from $\Delta = +0.190$ to $\Delta = +0.222$ under the judge, the opposite of what a string-level EM artifact would predict.

1 Introduction

Dynamic retrieval-augmented generation (RAG) systems retrieve external evidence during decoding to ground generation in source material (**???**). At the core of every such system sits a *retrieval trigger*: a scalar function of the partial generation that decides when to fire retrieval. We restrict our audit to *training-free* triggers. Prior work spans six signal families: attention-based DRAGIN (**?**), token-probability FLARE (**?**), entropy-trend ETC (**?**), hidden-state SeaKR (**?**) and INSIDE (**?**), multi-sample text-consistency RSUS (**?**), and hedging-marker SELFDOUBT-T1 (**?**). A contemporaneous answer-trajectory family tracks step-level dynamics of the model's own predicted answer (**???**). All seven families share the same evaluation convention: rank traces by the trigger score and report raw AUROC against a binary correctness label.

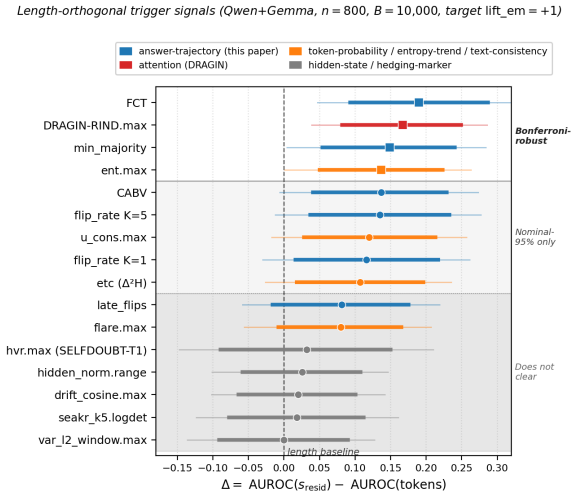


Figure 1: **Four trigger signals from three families carry length-orthogonal information beyond the length baseline; FCT (this paper) is strongest.** Forest plot of Δ = AUROC(s_{resid}) − AUROC(`tokens_generated`) on `lift_em`$= +1$ (n=800 paired traces). Thick bars: 95% paired-bootstrap CIs; thin extensions: Bonferroni-corrected 99.67% CIs at k=15. Setup: §**??**. Tables: App. **??**, **??**.

Recent work has shown this convention is length-confounded on adjacent QA selective-prediction and uncertainty-quantification (UQ) tasks. **?** prove that mutual length-bias on UQ scores and correctness functions systematically distorts AUROC. **?** show the empirical consequence: detection AUROC drops by up to 45.9% when switching from ROUGE to an LLM-as-Judge label. **?** attribute the underlying behavior to verbosity compensation on multi-hop QA, and **?** document a related correctness-function failure across approx-

imate metrics. The methodological response is **?**'s UNCERTAINTY-LINE, which regresses each UQ score on output length and uses the residual as a length-invariant estimate. The dynamic-RAG trigger pool is the missing application.

We extend the length-residualization protocol of **?** to the training-free dynamic-RAG trigger pool. Rather than score triggers against single-arm correctness, we re-decode each closed-book trace under gold context and ask whether retrieval flips a wrong answer to a right one, defining a paired-oracle target `lift_em`$=+1$ conceptually motivated by **?**. This isolates the *causal* retrieval-helpfulness signal from the length-correlated incorrectness signal that the published pool actually measures (§**??**). We also propose **FCT** (first-commitment time), a forward-looking variant of the answer-trajectory primitive (**????**). FCT reads K=5 majority answers from non-thinking-mode chat models, where neither Datta's `</think>`-anchored forced completion nor **?**'s finite-set continuation projection applies directly (App. **??**). On the held-out 2Wiki cells, length itself becomes anti-predictive of retrieval need, inverting the verbosity-compensation pattern **?** document on adjacent QA tasks.

Contributions. (1) A length-residualized audit of the dynamic-RAG trigger pool, the first to our knowledge, scored against a paired-oracle target (`lift_em`$= +1$). (2) **FCT** (first-commitment time), our forward-looking variant of the answer-trajectory primitive and the strongest length-orthogonal trigger in the audited pool. (3) A correctness-function robustness check against **?** and **?**, under which the survivor matrix is preserved and FCT *strengthens*.

2 Related Work

The length-confound critique of standard UQ AUROC evaluation comes from QA UQ and hallucination detection (**??????**). The answer-trajectory family (**???**) provides the parent primitives for FCT. Two related supervised methods sit outside our training-free scope: **?** train hidden-state probes for streaming hallucination detection, and **?** contrastively shape answer embeddings under latent perturbation at the trace boundary, where their "answer agreement" differs from our K=5 boundary-completion "answer-trajectory." Trained adaptive-retrieval policies likewise fall outside our audit (**??**). Our paired-oracle target operationalizes the question of when retrieval helps a parametric model (**?**), replacing static entity-popularity proxies with per-trace closed-vs-oracle re-decode.

3 Methods

3.1 Paired-Oracle `lift_em` Protocol

For each question q, we run two greedy decodes on the same chat model: a closed-book pass that produces trace $\tau_c^{(q)}$ and answer $a_c^{(q)}$, and an oracle pass conditioned on the gold supporting paragraphs that produces $a_o^{(q)}$. We read all trigger signals from $\tau_c^{(q)}$. The oracle decode only supplies the label. The discrimination target is

$$\texttt{lift_em}(q) = \mathbb{1}\Big[\mathrm{EM}(a_o^{(q)}) > \mathrm{EM}(a_c^{(q)})\Big], \quad (1)$$

which flags traces where retrieval would convert a wrong answer to right. The published trigger pool instead targets single-arm correctness, which length predicts moderately well on multi-hop QA (§**??**).

We evaluate on HotpotQA (**?**) and 2WikiMultiHopQA (**?**), holding the latter out as a cross-dataset cell, at n=200 paired traces per architecture-by-dataset cell. The two models are Qwen2.5-7B-Instruct (dense) and Gemma-4-26B-A4B-it (MoE-hybrid), chosen for cross-architecture coverage. The positive class is rare: 108 of 800 paired traces, base rate 13.5%.

3.2 Length Residualization

We adopt the residualization protocol of **?**. For each trigger signal s on a cohort of paired traces, we fit OLS of s on $\log(\texttt{tokens}+1)$ and evaluate AUROC on the residual:

$$s_{\text{resid}} = s - \hat{\alpha} - \hat{\beta}\,\log(\texttt{tokens}+1). \quad (2)$$

$\mathrm{AUROC}(s_{\text{resid}})$ then measures whether the signal carries information beyond length. Our $\log(\texttt{tokens}+1)$ parameterization agrees within $\varepsilon \leq 0.029$ AUROC with the public reference implementation on our cohorts (App. **??**). Several signals in the audited pool (ETC, FLARE, SeaKR, SELFDOUBT-T1) are originally online or per-token, and we evaluate each as a trace-level summary statistic with granularity adapters in Appendix **??**.

3.3 FCT and the Answer-Trajectory Family

At each reasoning-step boundary (detected by tokenizer-level sentence-end token matchings), we

sample $K{=}5$ boundary completions at temperature $\tau{=}0.6$, top-p 0.9, and extract a per-step answer from each via a regex over the model's `Answer:` prefix. From the resulting per-step $K{=}5$ majority sequence $\mathcal{M} = (m_1, \ldots, m_n)$ we compute four trace-level aggregations:

- `flip_rate`: fraction of consecutive-step majority disagreements;
- `late_flips`: count of $K{=}5$ answer-*set* disagreements in the last half of the trace;
- `min_majority`: $1 - \min_t(\max_{\text{count}}/K)$, the worst within-step disagreement;
- **FCT**, a forward-looking first-commitment time

$$\text{FCT} = \frac{\min\{i : m_j = a_{\text{final}}\ \forall\, j \geq i\}}{\max(1, n-1)} \in [0, 1]. \tag{3}$$

High FCT denotes late convergence: the trace has not settled on its final answer until close to the end of the decode. All four aggregations adapt the `AnswerSwitches` primitive of **?** to non-thinking-mode chat models with $K{=}5$ majority-answer extracts. Neither Datta's `</think>`-anchored forced completion nor **?**'s finite-set continuation projection applies directly in this regime (App. **??**).

CABV (continuous answer-belief velocity). CABV is a continuous between-step generalization of `flip_rate`: at each populated step we build a Laplace-smoothed empirical distribution over the trace's $K{=}5$ answers and average the Jensen–Shannon divergence between consecutive steps. CABV reduces to `flip_rate` when each step concentrates on a single answer, and is closest in spirit to the inference-time instability signal of **?**, defined on JSD between token distributions rather than $K{=}5$ answer distributions.

4 Results

4.1 Cross-Architecture Survivor Matrix

Of the 16 audited signals, nine clear the length baseline at nominal 95% paired-bootstrap, and only four survive Bonferroni correction at $k{=}15$ (Figure **??**): attention-based DRAGIN-RIND ($\Delta = +0.167$), token-probability `ent.max` ($+0.137$), answer-trajectory `min_majority` ($+0.149$), and **FCT** ($+0.190$, the strongest length-orthogonal trigger in the pool). The four survivors span three of the seven declared families, which suggests the length-orthogonal signal is not a quirk of any one mechanism. Two families have no nominal-95% survivor: hidden-state (SeaKR, INSIDE, the CRV [Concept Representation Vector] variants) and hedging-marker (SELFDOUBT-T1) are entirely absorbed by length under our protocol, consistent with their high length-correlation per cell (App. **??**). The full table is in App. **??**.

4.2 Length Is Anti-Predictive on the Held-Out 2Wiki Cells

The Bonferroni-robust four answer one question: which signals beat length? A sharper question is when length itself stops being informative. On the two held-out 2Wiki cells, raw answer length is at or below chance for predicting retrieval need: `tokens_generated` reaches AUROC **0.457** on Qwen $\times$ 2Wiki and **0.518** on Gemma $\times$ 2Wiki. The verbosity-compensation pattern **?** document on the same LongBench bundle does not transfer to `lift_em`: longer multi-hop traces are no longer reliably retrieval-helpful. The four Bonferroni-robust survivors clear the length baseline by a wider margin on these cells than on HotpotQA, meaning the raw-AUROC protocol miscounts the length signal on exactly the cells where retrieval matters most.

4.3 FCT Discriminates Strongest on the Held-Out MoE Cell

The cells where length fails are the cells where FCT succeeds. On Gemma $\times$ 2Wiki, the cell with the worst length proxy (`tokens_generated` 0.518), FCT reaches raw AUROC **0.784**, a $+0.266$ absolute gap over length (Table **??**). The same pattern repeats on Qwen $\times$ 2Wiki: where the length proxy is below chance (0.457), FCT's margin ($+0.147$) is wider than on either HotpotQA cell. The answer-trajectory primitive carries the most independent information where the length proxy collapses.

Cell	n	helped	FCT AUROC
Qwen $\times$ HotpotQA	200	31	0.677
Qwen $\times$ 2Wiki	200	34	0.604
Gemma $\times$ HotpotQA	200	19	0.642
Gemma $\times$ 2Wiki	200	24	**0.784**

Table 1: Per-cell raw AUROC of FCT on `lift_em`$=+1$.

4.4 Survivor Set Is Robust to a Semantic-Equivalence Judge

The natural alternative explanation for the survivor matrix is that it reflects EM's string-matching artifact rather than a length-orthogonal signal. To rule this out we relabelled both arms of every paired trace via a GPT-4o-Mini judge (**?**'s exact choice) and re-ran the audit, addressing the correctness-function critique of **?** and **?**. EM and the judge agree at Cohen's $\kappa = 0.707$ (substantial agreement; **?**). Eight of the nine paired-significant survivors under EM remain significant under the judge, with a single swap (`etc` loses, `flare.max` gains) that leaves the Bonferroni-robust four intact. FCT itself *strengthens* under the judge from $\Delta = +0.190$ to $\Delta = +0.222$, the opposite of what a string-level EM artifact would predict under semantic relabeling (App. **??**).

4.5 The $K{=}5$ Multi-Sample Variant Is Null

A second design choice worth stress-testing is the $K{=}5$ sampling regime in the answer-trajectory family. On both architectures, the paired-bootstrap CI on $\Delta_{K=5-K=1}$ for `flip_rate` straddles zero (Qwen -0.002, $[-0.040, +0.031]$; Gemma $+0.033$, $[-0.025, +0.092]$; App. **??**). The audit reports $K{=}5$ throughout because that was the capture protocol when the ablation came back null, and for deployment $K{=}1$ delivers the same triggering performance at $5\times$ less compute per boundary.

5 Discussion

5.1 Why Length Predicts Retrieval-Helpfulness

On the secondary `closed_em`$=0$ target, `tokens_generated` reaches raw AUROC 0.539 on the headline pool, competitive with several internal signals. This pattern matches **?**'s verbosity-compensation mechanism. A plausible causal chain is length $\rightarrow$ verbosity compensation $\rightarrow$ uncertainty $\rightarrow$ incorrectness $\rightarrow$ retrieval need, and length predicts `closed_em`$=0$ moderately well because it proxies the upstream incorrectness step. The paired-oracle target `lift_em`$= +1$ strips that step by conditioning on closed-book wrongness, which is why length collapses on the held-out 2Wiki cells (§**??**). Residualization (§**??**) provides the complementary control on any candidate trigger.

5.2 Why Answer-Trajectory Stability Predicts Retrieval Need

The mechanism behind FCT reads naturally as *premature commitment*, the interpretation **?** give to their `AnswerSwitches` and $t^\star$ primitives. The model decides on an answer early and continues to verbalize reasoning that supports the commitment, with most subsequent steps as post-decision tail. **?** apply this observation to early stopping: if the model has decided, no further compute is needed. Retrieval triggering reads the same primitive in the opposite causal direction. The traces where the model has not decided, where the answer trajectory continues to flip through the reasoning, are exactly the traces where external evidence would resolve the uncertainty. The paired-oracle target `lift_em`$= +1$ formalizes that question directly. The same primitive supports two complementary operations: detect stabilization to *stop*, detect non-stabilization to *retrieve*.

6 Conclusion

We presented the first length-residualized audit of the dynamic-RAG trigger pool, extending the protocol of **?** to a paired-oracle target (`lift_em`$= +1$). On two open-weight chat models over HotpotQA and 2WikiMultiHopQA, nine signals beat the length baseline at 95% paired-bootstrap and four survived Bonferroni at $k{=}15$ across three families. The strongest length-orthogonal trigger is **FCT**, our forward-looking variant of the answer-trajectory primitive, exceeding length by $\Delta = +0.190$ AUROC and reaching raw 0.784 on the held-out Gemma $\times$ 2Wiki cell. The survivor set is robust to a GPT-4o-Mini semantic-equivalence judge ($\kappa = 0.707$), with FCT itself strengthening under the judge. The same answer-trajectory primitive supports two complementary deployment operations: detect stabilization to *stop*, detect non-stabilization to *retrieve*.

Limitations

Signal-level audit, not end-to-end pipelines. We audit trigger *signals* as defined in their original papers, not full deployment pipelines (threshold tuning, retriever choice, prompt format). Our claims pertain to signal-level discriminative power on `lift_em`.

Bonferroni knife-edge. Two of the four Bonferroni-robust survivors (`ent.max`,

min_majority) carry ∼99.67% lower bounds within 5×10^{-3} of zero (App. **??**). The four-survivor count is therefore fragile to small perturbations. The FCT and DRAGIN-RIND lower bounds are robust at $+0.047$ and $+0.039$.

Architecture and dataset coverage. The headline pool covers two architectures and two multi-hop QA benchmarks. The cheapest extensions are a third open-weight chat model (e.g., Llama-3.1-8B-Instruct) and a third multi-hop benchmark (MuSiQue, **?**). The current FCT Bonferroni lower bound ($+0.047$) is the statistical floor additional cells would have to displace.

Dataset framing: held-out, not OOD. We treat 2WikiMultiHopQA as a *held-out* cell relative to HotpotQA, not as out-of-distribution in the strict sense: both are multi-hop QA over Wikipedia with comparable question complexity. The cell-dependent length behavior (§**??**) is a dataset-specific finding on multi-hop QA rather than a claim about OOD generalization, and a stricter cross-distribution evaluation is future work.

Correctness function. We use exact-match equality on both arms of the paired-oracle target. The paired design (§**??**) and the LLM-as-Judge robustness check (App. **??**) partially insulate the headline against this choice. Containment-match and F_1 variants remain future work.

Non-thinking-mode chat models. The audited models are non-thinking-mode chat variants. **?** developed their answer-trajectory primitive on thinking-mode reasoning models with explicit </think> tokens. Our $K=5$ boundary-completion adaptation is not byte-faithful to that protocol.

Risks. Our work concerns evaluation methodology for retrieval triggering on open-weight chat models and public multi-hop QA benchmarks. We release no new datasets or trained models, and direct risks are limited. One methodological risk is that the length-residualized protocol could be cited in isolation to defend deployed dynamic-RAG systems whose triggers track length more strongly than residualization alone would reveal. We therefore recommend reporting raw, length-residualized, and LLM-judge-validated AUROC together (§**??**) rather than substituting one for another.

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A Bonferroni-Corrected Survivor Set

Table **??** reports the Bonferroni-corrected ∼99.67% paired-bootstrap CI of $\Delta = \mathrm{AUROC}(s_{\mathrm{resid}}) - \mathrm{AUROC}(\texttt{tokens_generated})$ at $k{=}15$ ($\alpha = 0.05/15 = 0.0033$) on the headline pool. Bonferroni k is the number of signals tested for the survivor-matrix claim. Four signals retain a strictly positive lower CI bound under correction: FCT, DRAGIN-RIND, `min_majority`, and `ent.max`.[1]

Signal	Δ	95% CI	99.67% CI (Bonf.)
Bonferroni-robust:			
FCT	+0.190	[+0.093, +0.287]	[+0.047, +0.329]
DRAGIN-RIND	+0.167	[+0.082, +0.249]	[+0.039, +0.286]
min_majority	+0.149	[+0.054, +0.240]	[+0.004, +0.284]
ent.max	+0.137	[+0.050, +0.223]	[+0.001, +0.263]
Nominal-95% only:			
CABV	+0.137	[+0.041, +0.229]	[−0.006, +0.273]
flip_rate $K{=}5$	+0.135	[+0.037, +0.233]	[−0.012, +0.277]
u_cons.max	+0.120	[+0.028, +0.213]	[−0.017, +0.257]
flip_rate $K{=}1$	+0.116	[+0.016, +0.217]	[−0.030, +0.261]
etc ($\Delta^2 H$)	+0.107	[+0.018, +0.196]	[−0.026, +0.235]

Table 2: Bonferroni-corrected survivor set on the headline pool (Qwen+Gemma, $n{=}800$ paired, `lift_em`$= +1$, $B{=}10{,}000$). The `etc` ($\Delta^2 H$) signal is the Entropy-Trend Constraint trigger of **?**.

B Full Headline Survivor Matrix

Table **??** reports all 16 audited signals on the headline pool with length-residualized AUROC and Δ vs. `tokens_generated`.

C Correctness-Function Robustness: LLM-as-Judge

Protocol. For each of the $n{=}800$ paired traces in the headline pool, we re-labelled both arms via an OpenAI `gpt-4o-mini` semantic-equivalence judge under a strict three-sentence system prompt: "You are a strict semantic-equivalence judge for multi-hop QA. Your task is to decide whether a model-predicted answer is semantically equivalent to a gold answer. Reply with exactly one word: YES or NO. No explanation." The judge target is

$$\texttt{lift_em}_{\mathrm{JUDGE}}(q) = \mathbb{1}\big[\mathrm{judge}(a_o^{(q)}){=}\mathrm{YES} \wedge \mathrm{judge}(a_c^{(q)}){=}\mathrm{NO}\big]. \quad (4)$$

We re-ran the full headline survivor matrix with paired bootstrap $B{=}10{,}000$.

Inter-label agreement. EM identifies 108 helped traces and the judge identifies 113; the intersection is 87. The two labels jointly produce a multi-class Cohen's $\kappa = 0.707$ on the 3-class `lift_em` $\in \{+1, 0, -1\}$ label set, in the substantial-agreement band of **?**.

Survivor matrix transfer. Table **??** reports the headline survivor matrix under both correctness functions side-by-side. Eight of the nine paired-significant survivors under EM remain significant under the judge, with a single swap: `etc` loses significance and `flare.max` gains it. FCT strengthens under the judge from $\Delta = +0.190$ [+0.093, +0.287] to $\Delta = +0.222$ [+0.132, +0.312] (+17% in point estimate, +42% in 95% CI lower-bound margin), the opposite of what a string-level EM artifact would predict. All four Bonferroni-robust survivors retain nominal-95% significance under the judge.

Caveat. **?** document that GPT-4-tier judges exhibit a verbosity bias, preferring longer responses; our cross-judge check demonstrates *pattern stability* rather than full verbosity-independence. The load-bearing length-orthogonality argument rests on the UNCERTAINTY-LINE residualization (§**??**) and the paired-oracle design (§**??**); the judge check confirms the survivor matrix is not an artifact of any single correctness function.

D Uncertainty-Line Reference Replication

To pre-empt the reimplementation critique, we ran our protocol (Eq. **??**) and the **?** reference implementation side-by-side on the same trace cohort, decomposing the comparison into two ε components. $\varepsilon_{\mathrm{protocol}}$ isolates the math difference (our $\log(\texttt{tokens}+1)$ single-cohort OLS against the reference MinMax-scaled linear regression on the

[1] $k{=}15$ was the multiple-comparison count fixed at test time, when the audited pool included a CABV×LAAS composite that we later dropped from presentation (§**??**). The current table reports 16 audited signals plus the length baseline. The Bonferroni-corrected CI half-widths in the table (≈ 0.12–0.14 AUROC at $B{=}10{,}000$) dominate any plausible k-sensitivity at this signal pool size. We report the test as run.

Signal	Family	Resid AUROC	Δ vs. tokens	95% CI
tokens_generated	length	0.531	—	(baseline)
FCT	answer-trajectory	0.691	+0.190	[+0.093, +0.287]
DRAGIN-RIND	attention	0.640	+0.167	[+0.082, +0.249]
min_majority	answer-trajectory	0.626	+0.149	[+0.054, +0.240]
ent.max	token-probability	0.611	+0.137	[+0.050, +0.223]
CABV	answer-trajectory	0.641	+0.137	[+0.041, +0.229]
flip_rate K=5	answer-trajectory	0.640	+0.135	[+0.037, +0.233]
u_cons.max	K=3 text	0.593	+0.120	[+0.028, +0.213]
flip_rate K=1	answer-trajectory	0.621	+0.116	[+0.016, +0.217]
etc ($\Delta^2 H$)	entropy-trend	0.580	+0.107	[+0.018, +0.196]
late_flips	answer-trajectory	0.586	+0.081	[−0.016, +0.175]
flare.max	token-probability	0.553	+0.080	[−0.008, +0.165]
hvr.max (SELFDOUBT-T1) (**?**)	hedging-marker	0.505	+0.032	[−0.089, +0.150]
hidden_norm.range	hidden (CRV)	0.499	+0.026	[−0.058, +0.108]
drift_cosine.max	hidden (CRV)	0.493	+0.020	[−0.064, +0.101]
seakr_k5.logdet	K=5 hidden	0.491	+0.018	[−0.077, +0.112]
var_l2_window.max	hidden (CRV)	0.473	+0.000	[−0.091, +0.090]

Table 3: Full cross-architecture survivor matrix on the headline pool (Qwen2.5-7B + Gemma-4-26B-A4B-it × {HotpotQA, 2WikiMHQA}, n=800 paired traces). All AUROCs are length-residualized (§**??**). 95% CIs are paired-bootstrap (B=10,000) of $\Delta = \mathrm{AUROC}(s_{\text{resid}}) - \mathrm{AUROC}(\texttt{tokens_generated})$. **Bold**: Bonferroni-robust at k=15 (lower bound of the 99.67% CI strictly positive; see Table **??**).

Signal	Δ_{EM}	Δ_{JUDGE}
FCT	+0.190	**+0.222**
DRAGIN-RIND	+0.167	+0.157
min_majority	+0.149	+0.155
ent.max	+0.137	+0.125
CABV	+0.137	+0.177
flip_rate K=5	+0.135	+0.178
u_cons.max	+0.120	+0.157
flip_rate K=1	+0.116	+0.171
etc ($\Delta^2 H$)	+0.107	+0.088*
flare.max	+0.080*	+0.115

Table 4: Side-by-side Δ under EM and LLM-as-Judge on the headline pool (n=800, B=10,000). *95% CI straddles zero. FCT *strengthens* under the judge. Eight of nine paired-significant EM survivors are preserved, and all four Bonferroni-robust survivors retain nominal-95% significance.

same cohort); $\varepsilon_{\text{sampling}}$ isolates train/test-split sampling noise at n=200 (Table **??**).

E FCT vs. Concurrent Commitment-Time Primitives: Operational Differences

FCT (§**??**) is motivated by the final-switch index $t^\star = \max\{i : A_i \neq A_{i+1}\}$ of **?**, with a concurrent forward-looking commitment-time primitive proposed by **?** on the model's finite-answer continuation distribution. The three are operationally distinct on five axes (Table **??**).

Arch	$\varepsilon^{\text{prot}}_{\max}$	$\varepsilon^{\text{prot}}_{\text{mean}}$	$\varepsilon^{\text{prot}}_{\text{med}}$	$\varepsilon^{\text{samp}}_{\max}$
Qwen2.5	0.029	0.012	0.012	0.111
Gemma	0.027	0.0095	0.0047	0.106

Table 5: $\varepsilon_{\text{protocol}} = |\text{ours}_{\text{resid}} - \text{LINE}_{\text{full}}|$; $\varepsilon_{\text{sampling}} = |\text{LINE}_{\text{full}} - \text{LINE}_{\text{split}}|$ (latter averaged over 3 seeds). Both architectures show $\varepsilon_{\text{protocol}} \leq 0.029$, well below the conventional threshold of 0.05. Train/test sampling noise ($\varepsilon_{\text{sampling}}$) dominates the protocol gap by a factor of 3–4×, indicating that the $\log(\texttt{tokens} + 1)$ vs. raw-MinMax math difference is small relative to evaluation-set variance at n=200.

F Length-vs-Signal Spearman ρ per Cell

Table **??** reports Spearman ρ between tokens_generated and each signal per (arch × dataset) cell on closed-book traces, the length-confound diagnostic of **?** (their Table 2). $|\rho| \geq 0.5$ indicates a strong length confound on the signal.

G K=5 vs. K=1 Ablation

Table **??** reports the paired-bootstrap $\Delta_{K=5-K=1}$ for flip_rate on lift_em= + 1, pooled per architecture. Both CIs straddle zero (§**??**).

H Adapter Notes

We audit signal-level discriminative power on lift_em (§**??**). Several primitives in the published trigger pool are originally framed as online per-token triggers or per-trace scores, and we evaluate

Axis	Datta $t^\star$	Zhang τ_{commit}	FCT (ours)
Temporal direction	Backward (last switch)	Forward (first stable)	Forward (first stable)
Signal class	Per-step answer	Continuation log-odds	$K{=}5$ majority answer
Sampling regime	$K{=}1$ forced-completion	Single deterministic state	$K{=}5$ stochastic
Application target	Early stopping	Pre-verbalization commitment	Retrieval triggering
Trace model class	Thinking (`</think>`)	Delayed-verdict templates	Non-thinking chat

Table 6: Operational differences across three concurrent commitment-time primitives. **?**'s $t^\star$ is backward-looking on per-step intermediate answers, applied to early stopping. **?**'s τ_{commit} is forward-looking on the model's finite-answer continuation distribution (log-odds in binary tasks), applied to pre-verbalization commitment analysis. FCT (ours) is forward-looking on $K{=}5$-majority answer extracts, applied to retrieval triggering. For monotonically converging trajectories, $t_{\text{commit}} \approx t^\star + 1$; under $K{=}5$ majority sampling on non-thinking-mode chat models this equivalence does not hold in general.

their underlying statistic at the trace level. The per-signal notes below state the original primitive, our trace-level adaptation, and any sign or granularity drift.

ETC (?). The original ETC operates on a token-level entropy stream; we evaluate $\Delta^2 H$ on the per-step mean entropy series captured at boundary-detection events (§**??**). We score $\max|\Delta^2 H|$ rather than the signed second-difference, so the adapter is magnitude-faithful but not sign-faithful.

FLARE (?). The original FLARE is an online per-token retrieval trigger ($\max_t p_t < \tau$); we audit the underlying primitive (low-confidence detection) as a trace-level summary $\max$ over steps of $1 - \min p_t$. The trigger *policy* is not audited; the trigger *statistic* is.

SeaKR (?). We use $K{=}5$ EOS hidden states for compute efficiency, smaller than the paper's reported best-k interval $K \in [10, 25]$. The regularized Gram determinant we compute is the same primitive as the paper's self-aware uncertainty score. INSIDE (**?**) is the closest peer, using the EigenScore on the centered sample covariance; the K nonzero eigenvalues of our uncentered Gram and INSIDE's centered covariance are rank-equivalent up to centering.

SELFDOUBT-T1 (?). HVR $= h/(v{+}1)$ is computed per step and aggregated as $\max$ over steps; the original paper computes a single trace-level HVR with a Tier-1 "HVR=0" high-precision correctness gate. The per-step adapter is recoverable as the paper's gate by setting our threshold to 0.

Signal	Q×Hot	Q×Wiki	G×Hot	G×Wiki
DRAGIN-RIND	+0.40	+0.35	+0.38	+0.38
ent.max	+0.31	+0.26	+0.42	+0.32
etc ($\Delta^2 H$)	+0.32	+0.36	+0.46	+0.40
flare.max	+0.39	+0.35	+0.45	+0.32
u_cons.max	+0.40	+0.31	+0.37	+0.39
seakr_k5	+0.37	+0.44	+0.46	+0.43
drift_cosine.max	+0.20	+0.18	−0.07	−0.01
var_l2_window.max	+0.19	+0.23	+0.22	−0.00
hidden_norm.range	+0.24	+0.29	+0.36	+0.16
hvr.max	n/a*	+0.06	+0.30	**+0.51**
flip_rate $K{=}5$	+0.22	+0.12	+0.21	+0.35
flip_rate $K{=}1$	+0.17	+0.15	+0.27	+0.21
late_flips	+0.29	+0.37	+0.25	+0.30
min_majority	+0.18	+0.12	+0.14	+0.09
CABV	+0.22	+0.17	+0.22	+0.36
FCT	+0.20	+0.14	+0.15	+0.30

Table 7: Spearman $\rho(\texttt{tokens_generated}, \text{signal})$ per cell on closed-book traces. Bold: $|\rho| \geq 0.5$. The peer-pool signals (DRAGIN, ent.max, etc, flare.max, u_cons.max, seakr_k5) carry $\rho \in [+0.26, +0.46]$, `hvr.max` reaches $+0.51$ on Gemma×2Wiki, and the auxiliary CRV diagnostics span $\rho \in [-0.07, +0.36]$. The answer-trajectory aggregations (bottom block) average ≈ 0.15 lower in $|\rho|$ than the peer-pool block, the per-cell quantitative version of the family-level finding from §**??**. *Qwen×Hot `hvr.max` is constant, so Spearman is undefined.

Arch	n	helped	$\Delta_{K=5-K=1}$	95% CI
Qwen	305	47	−0.002	$[-0.040, +0.031]$
Gemma	394	42	+0.033	$[-0.025, +0.092]$

Table 8: $K{=}5$ vs. $K{=}1$ `flip_rate` ablation on `lift_em`$= +1$, pooled per architecture.

Concurrent commitment-time primitives. Operational differences between **?**'s $t^\star$, **?**'s τ_{commit}, and our FCT are tabulated in Appendix **??**.

I Reproducibility

All trace JSONLs are committed at `runs/v2_headline_{arch}/` for arch ∈ {qwen2.5-7b-instruct, gemma-4-26b-a4b-it}. Slice manifests with SHA-256 checksums are at `runs/v2_headline_{arch}/checkpoints/slice_manifest.json` and decoding command lines are at `runs/v2_headline_{arch}/logs/00_master.log`. Analysis scripts are at `analysis/r1_paired_audit.py`, `analysis/a1_k1_vs_k5_ablation.py`, `analysis/uncertainty_line_audit.py`, `analysis/moonshot_explore.py`, `analysis/llm_judge_audit.py`, and `analysis/e7_bonferroni_and_spearman.py`

(the last produces Table **??** and Table **??**). The judge cache contains 1,600 per-call JSONs (oracle and closed arm for each of 800 paired traces) at `analysis/llm_judge_cache/`. Compute box: single NVIDIA A800 80 GB, wall-clock $\sim$15 h.